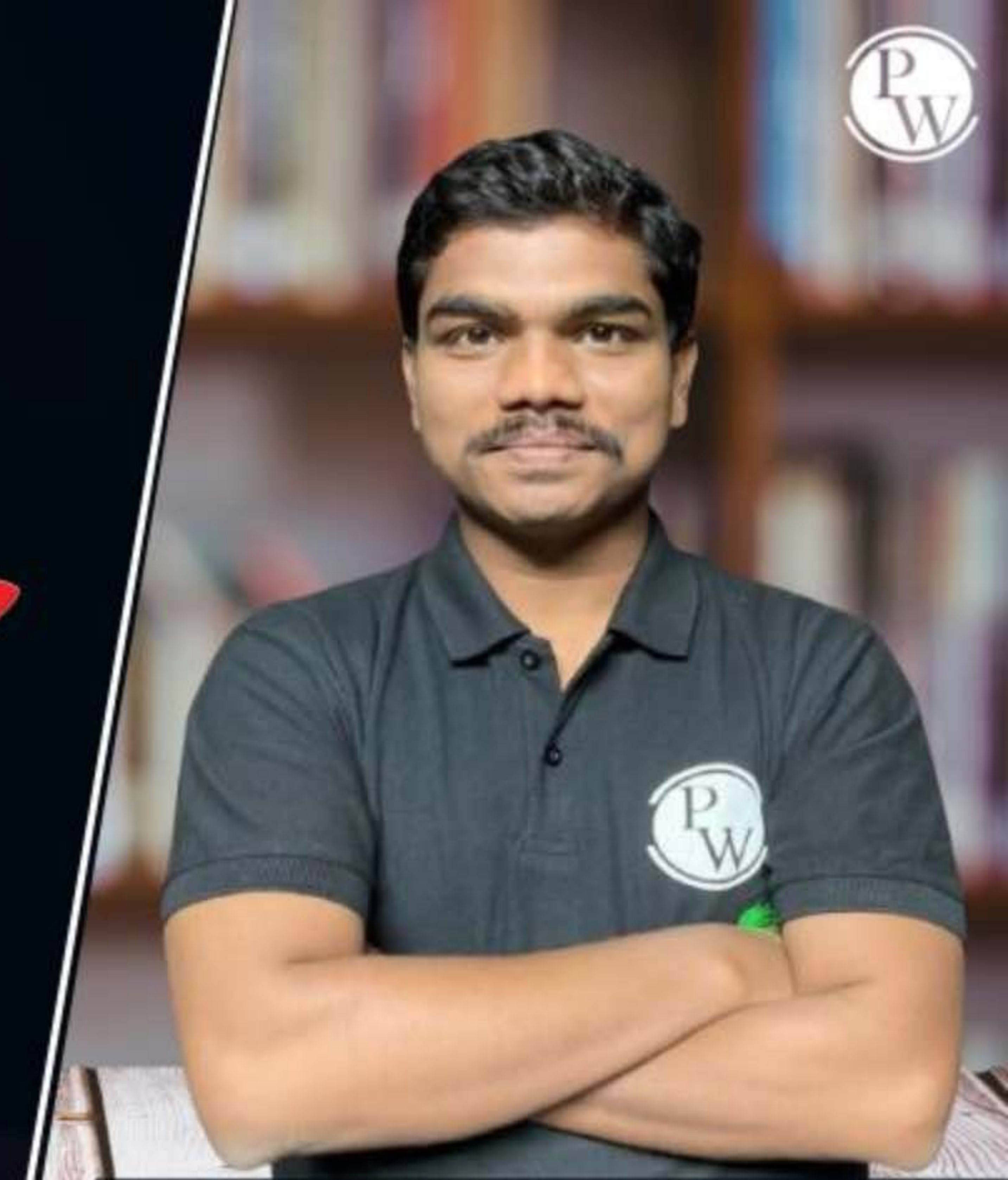
GATE- ALL BRANCHES
ENGINEERING MATHEMATICS



Linear Algebra

Lecture No. 4

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[Recap of previous lecture]

$\rightarrow$ Determinants.

$\rightarrow$ Properties of Determinants.

a) $|AB| = |A| \cdot |B| = |BA|$

b) $|kA_{n\times n}| = k^n \cdot |A|$

Where $k \in \mathbb{R}$.

$\rightarrow$ $|\text{adj}(A)| = |A|^{n-1}$.

$|\text{adj}(\text{adj}(A))| = |A|^{(n-1)^2}$.

Topics to be Covered

- **Topic** Rank of a Matrix
- **Topic** Linear Combination of Vectors
- **Topic** Vector Spaces
- **Topic** Linear Dependency and Independency

→ If two rows/columns of a determinant are identical/proportional then the value of the determinant is zero.

i.e

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ \alpha a_{11} & \alpha a_{12} & \alpha a_{13} \end{vmatrix} = \alpha \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \end{vmatrix} = 0.$$

$$|A| = a_{21} \times (-1)^{2+1} \underbrace{\begin{vmatrix} a_{12} & a_{13} \\ a_{12} & a_{13} \end{vmatrix}}_{0} + a_{22} \times (-1)^{2+2} \cdot \underbrace{\begin{vmatrix} a_{11} & a_{13} \\ a_{11} & a_{13} \end{vmatrix}}_{0}$$

$$+ a_{23} \cdot (-1)^{2+3} \cdot \underbrace{\begin{vmatrix} a_{11} & a_{12} \\ a_{11} & a_{12} \end{vmatrix}}_{0} = 0$$

→ If a row/column of a Matrix is zero, then the determinant of that Matrix is zero.

Ex: i.e $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \Rightarrow |A| = 0 \times (-1)^{2+1} \cdot \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + \cdots + 0 \times (-1)^{2+3} \cdot \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}$

$= 0.$

→ The determinant of a Matrix doesn't change by Performing Row/column Operations.

$$R_i \rightarrow R_i - \sum_{\substack{j=1 \\ j \neq i}}^{n} a_j R_j \qquad (\text{Ex: } R_3 \rightarrow R_3 - (R_1 - 2R_2))$$

Ex: Say $A = \begin{bmatrix} 12 & 3 & 4 \\ 7 & 8 & 9 \\ 19 & 11 & 13 \end{bmatrix}$

Consider $|A| = \begin{vmatrix} 12 & 3 & 4 \\ 7 & 8 & 9 \\ 19 & 11 & 13 \end{vmatrix}$

$R_3 \to R_3 - (R_1 + R_2)$

$|A| = \begin{vmatrix} 12 & 3 & 4 \\ 7 & 8 & 9 \\ 0 & 0 & 0 \end{vmatrix} = 0$

→ If sum of $(n-1)$ rows/columns is equal to n^{th} row/column, then that determinant is zero.

Rank of a matrix

A matrix '$A_{m\times n}$' is said to have a rank 'r' where $r \le \min\{m, n\}$.

if a) All Minors of 'A' of order $(r+1)\times(r+1)$ and above are zeroes.

b) $\exists$ atleast one Non-zero minor of order '$r\times r$'. It is denoted as $\rho(A) = r$.

Ex: $A = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 2 & 3 & 2 & 4 \\ 2 & 4 & 2 & 2 \end{bmatrix}_{3\times 4}$

considering 3×3 Minors:

$\begin{vmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ 2 & 4 & 2 \end{vmatrix} = 0$; $\begin{vmatrix} 1 & 2 & 1 \\ 2 & 3 & 4 \\ 2 & 4 & 2 \end{vmatrix} = 0$; $\begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 4 \\ 2 & 2 & 2 \end{vmatrix} = 0$; $\begin{vmatrix} 2 & 1 & 1 \\ 3 & 2 & 4 \\ 4 & 2 & 2 \end{vmatrix} = 0$

2×2 Minors:

$\begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} = -1 \neq 0$; ... 17 other (2×2) Minors.

$\Rightarrow \boxed{\rho(A) = 2}$

→ Row-echelon form of a Matrix:

A Matrix 'A'$_{m\times n}$ is said to be in row echelon form if

a) No. of zeroes before the 1st Non-zero element in any row is strictly less than No. of such zeroes in it's succeeding row.

b) Zero rows (if any) should lie at the bottom of the Matrix.

Ex: $A = \begin{array}{c} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 2 & 7 & 8 \\ 0 & 0 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}_{4\times 4} \end{array} \begin{array}{l} \to 1^{st}\text{ row} \\ \to 2^{nd}\text{ row} \\ \to 3^{rd}\text{ row} \\ \to 4^{th}\text{ row} \end{array}$; $A = \underbrace{\begin{bmatrix} 0 & 2 & 4 & 3 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}_{3\times 4}}_{\to \text{Row-echelon form.}}$

Ex: $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 4 \end{bmatrix} \longrightarrow$ Echelon form

Ex: $A = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & 2 & 0 & 4 \\ 0 & 0 & 2 & 1 \end{bmatrix} \longrightarrow$ Not in Row-echelon form.

How to convert a given Matrix into Row-echelon form?

Ans: Row Transformation / Row Operations

$\rightarrow$ Ex:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

$(a_{11} \neq 0)$

$$R_i \rightarrow R_i - \left(\frac{a_{i1}}{a_{11}}\right) R_1 \quad \forall\, i = 2 \text{ to } n.$$

$$A \sim \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a'_{22} & a'_{23} & \cdots & a'_{2n} \\ 0 & a'_{32} & a'_{33} & \cdots & a'_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & a'_{m2} & a'_{m3} & \cdots & a'_{mn} \end{bmatrix}$$

$$R_i \rightarrow R_i - \left(\frac{a'_{i2}}{a'_{22}}\right) \cdot R_2 \quad \forall\, i = 3 \text{ to } n.$$

Pivot elements

$$A \sim \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a'_{22} & a'_{23} & \cdots & a'_{2n} \\ 0 & 0 & a''_{33} & \cdots & a''_{3n} \\ 0 & 0 & a''_{43} & \cdots & a''_{4n} \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & a''_{m3} & \cdots & a''_{mn} \end{bmatrix}$$

$\downarrow$

Row-echelon form

(Finally).

Ex:

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 4 & 0 & 8 & 4 \\ 3 & 6 & 9 & 12 & 15 \\ 4 & 1 & 7 & 2 & 3 \end{bmatrix}_{4\times 5}$$

$R_2 \to R_2 - (2)R_1$

$R_3 \to R_3 - 3R_1$

$R_4 \to R_4 - 4R_1$

$$\Rightarrow A \sim \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & -6 & 0 & -6 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & -7 & -5 & -14 & -17 \end{bmatrix}$$

$R_3 \longleftrightarrow R_4$ (Interchange)

$$\Rightarrow A \sim \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & -6 & 0 & -6 \\ 0 & -7 & -5 & -14 & -17 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$R_2 \Longleftrightarrow R_3$ (swapping to eleminate zero pivot).

$$\Rightarrow A \sim \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & -7 & -5 & -14 & -17 \\ 0 & 0 & -6 & 0 & -6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(Row-echelon form).

$\rho(A) = 3$

If No. of Non-zero rows in Row-echelon form of a Matrix is 'r' then rank of that Matrix $= r$.

→ **Properties of Rank:**

① $\rho(A_{m\times n}) \leq \min(m, n)$

② $\rho(A) = \rho(A^n)$

③ $\rho(AB) \leq \min\{\rho(A), \rho(B)\}$.

Linear Combination of Vectors

If $\vec{x}_1, \vec{x}_2, \vec{x}_3, \ldots, \vec{x}_n$ are vectors in $\mathbb{R}^n$;

$\Rightarrow \vec{x} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix}_{n\times 1}$; then the combination $\sum c_i \vec{x}_i$ is called

($c_1, c_2, \ldots, c_n$ are scalars).

linear combination of vectors.

$\vec{x} = \begin{pmatrix} a \\ b \end{pmatrix}_{2\times 1} \Rightarrow \vec{x} \in \mathbb{R}^2$.

$\vec{x} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}_{3\times 1} \Rightarrow \vec{x} \in \mathbb{R}^3$.

$\Rightarrow$ linear combination is

$$c_1\vec{x}_1 + c_2\vec{x}_2 + \cdots + c_n\vec{x}_n$$

Linear Dependency and Independency

If a linear combination of vectors is defined by $\vec{X} = c_1\vec{x}_1 + c_2\vec{x}_2 + \cdots + c_n\vec{x}_n$ then the combination is said to be $\underbrace{\text{linearly Independent}}$ when $\vec{X} = \vec{0}$ only for $c_1 = c_2 = c_3 = \cdots = c_n = 0$.

Ex: Say $\vec{x}_1 = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$; $\vec{x}_2 = \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix}$; $\vec{x}_3 = \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix}$.

linear combination is $c_1\vec{x}_1 + c_2\vec{x}_2 + c_3\vec{x}_3 = \vec{x} \Rightarrow \vec{x} = \begin{pmatrix} 2c_1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 3c_2 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 4c_3 \end{pmatrix}$

$\Rightarrow \vec{x} = \begin{pmatrix} 2c_1 \\ 3c_2 \\ 4c_3 \end{pmatrix} = \vec{0} \Rightarrow \exists$ $c_1 = 0$; $c_2 = 0$; $c_3 = 0$ as only solution for $\vec{x} = \vec{0}$.

If $c_1\vec{x}_1 + c_2\vec{x}_2 + \cdots + c_n\vec{x}_n = \vec{0}$ if $c_1, c_2, \ldots c_n$ are not all zero simultaneously, then the Combination is Said to be linearly defendent.

Ex: $\vec{x}_1 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$; $\vec{x}_2 = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$; $\vec{x}_3 = \begin{pmatrix} 3 \\ 7 \\ 4 \end{pmatrix}$; $\vec{x}_4 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$

Linear combination $\vec{x} = c_1\vec{x}_1 + c_2\vec{x}_2 + c_3\vec{x}_3$.

$$\Rightarrow \vec{x} = c_1\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} + c_2\begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} + c_3\begin{pmatrix} 3 \\ 7 \\ 4 \end{pmatrix} = \begin{pmatrix} c_1 + c_2 + 3c_3 \\ c_1 + 3c_2 + 7c_3 \\ 2c_1 + c_2 + 4c_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

on solving $c_1 = 1$; $c_2 = 2$; $c_3 = -1$

$$\Rightarrow \vec{x} = \vec{x}_1 + 2\vec{x}_2 - \vec{x}_3 + 0\cdot\vec{x}_4 = \vec{0}. \Rightarrow \boxed{\vec{x}_3 = \vec{x}_1 + 2\vec{x}_2}$$

Consider $[\vec{x}_1, \vec{x}_2\ \vec{x}_3] = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & 7 \\ 2 & 1 & 4 \end{bmatrix}$ (say A).

$\Rightarrow \det(A) = 0$ $\quad\left(\because \vec{x}_3 = \vec{x}_1 + 2\vec{x}_2 \right.$

$\Rightarrow (\text{col } 3 \to \text{col } 3 - (\text{col } 1 + 2 \cdot \text{col } 2)))$

<u>Note</u>: If 'n' vectors $\vec{x}_1, \vec{x}_2, \ldots, \vec{x}_n$ where $x_i \in \mathbb{R}^n$ and they are linearly dependent,

$\det([\vec{x}_1\ \vec{x}_2 \ldots \vec{x}_n]) = 0$

If linearly independent, then determinant $\neq 0$.

2 mins Summary

→ Concept of Rank

→ Row-echelon form

→ Linear dependency & Independency of Vectors.

→ Probabilistic M/c learning by kevin. P. Murthy.

THANK - YOU